ATOMIC-LEVEL MATERIALS SIMULATION

Module Code: MES306TC

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Symmetry and Band Structure

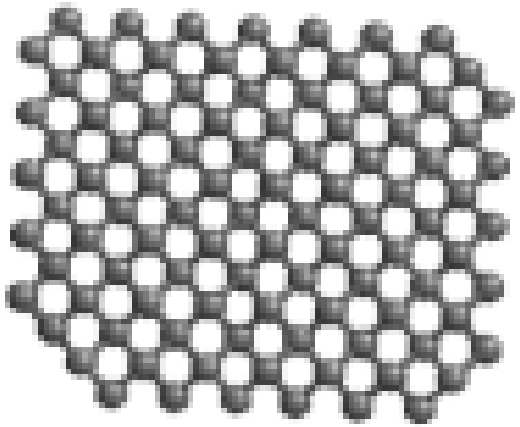
I. Crystal Symmetry

II. Band Structure

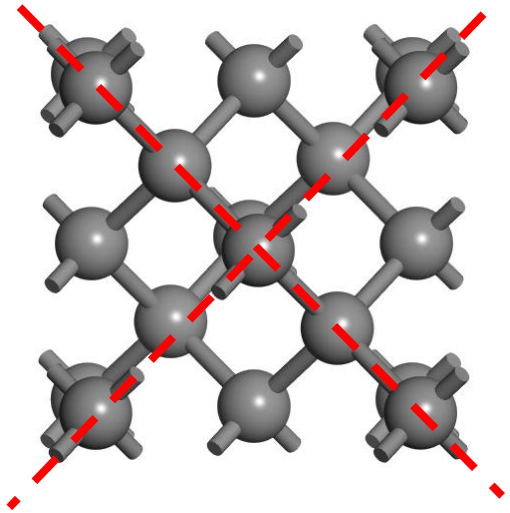
I. Crystal Symmetry

A 3D geometric matrix showing a single unit cell undergoing four distinct transformations: a rotation around an axis, a mirror reflection across a plane, an inversion through the center, and a translation

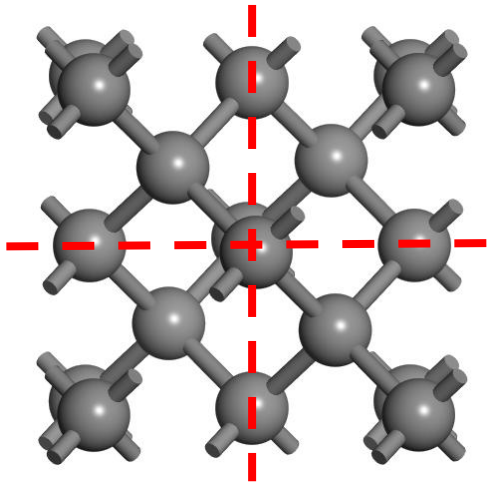
Diamond Crystal Structure



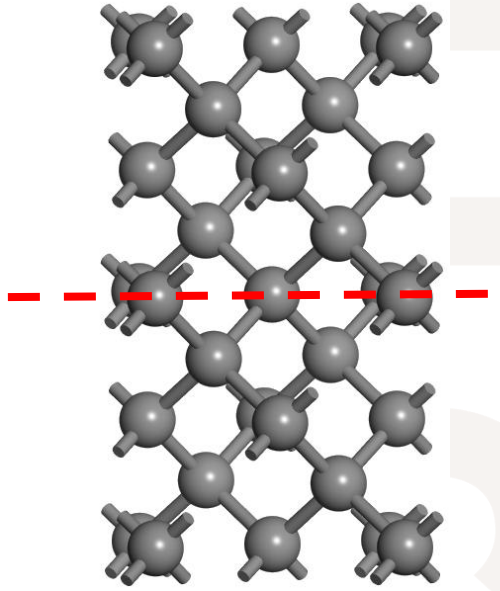
Rotation Symmetry



Mirror Reflection Symmetry



Inversion Symmetry



Translation Symmetry

I. Crystal Symmetry

The Symmetry Operator $\hat{S}$: A geometric transformation that moves the crystal lattice into an arrangement that is completely indistinguishable from its original state

Rotation (C_n): Rotating the crystal by $360^o/n$ (where, n = 1, 2, 3, 4, 6 are the only allowed values in periodic crystals)

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & cos\theta & -sin\theta \\ 0 & sin\theta & cos\theta \end{vmatrix}$$

x axis

$$\begin{vmatrix} cos\theta & 0 & sin\theta \\ 0 & 1 & 0 \\ -sin\theta & 0 & cos\theta \end{vmatrix}$$

y axis

$$\begin{vmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

z axis

I. Crystal Symmetry

Mirror Reflection (σ): Mirroring atoms across a specific lattice plane

$$\begin{vmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

***x* mirror**

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

***y* mirror**

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix}$$

***z* mirror**

I. Crystal Symmetry

Inversion (*i*): Mapping every atomic coordinate from *r* to -*r* through a central inversion point

$$\begin{vmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{vmatrix}$$

Space Groups: The combination of all point group operations plus translational operations (like glide planes and screw axes). There are exactly 230 unique Space Groups in 3D space

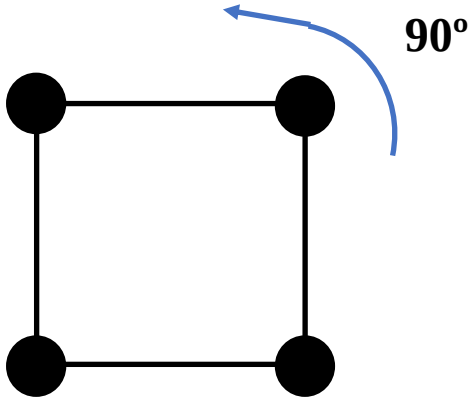
I. Crystal Symmetry

Invariance of the Hamiltonian: Degeneracy and Quantum Labels

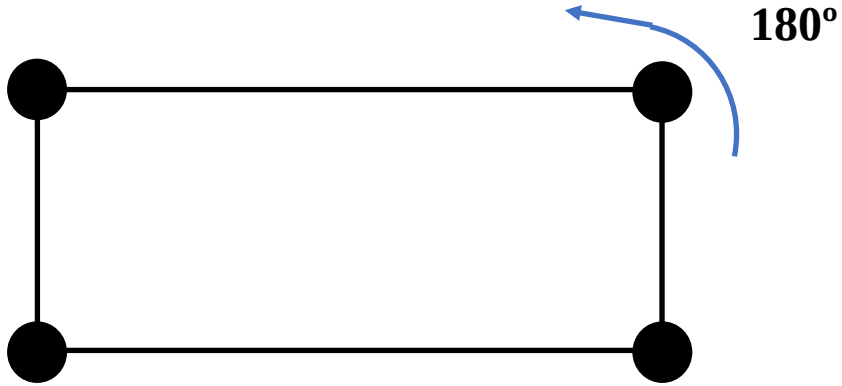
If a crystal possesses a symmetry operation $\hat{S}$, its quantum mechanical Hamiltonian H must commute with that operator

$$[H, S] = HS - SH = 0$$

Shared Eigenstates: The electronic wave functions Ψ_r can be chosen to be simultaneous eigenstates of both the energy Hamiltonian and the symmetry operators



C_4



C_2

I. Crystal Symmetry

Eigen Value Degeneracy: If two *k* points in the Brillouin Zone are related by a symmetry operation ($k' = Sk$), they must share the exact same energy eigenvalues

$$E_n(k') = E_n(Sk) = E_n(k)$$



High-symmetry points (like *Γ* or *K*) allow multiple bands to touch or cross without repelling each other

Because the Hamiltonian commutes with the symmetry of the crystal, symmetry acts as a rigid protector of electronic states.

For example, in Graphene, the famous Dirac cone—where the conduction and valence bands touch linearly—is strictly protected by a combination of inversion and time-reversal symmetry at the *K* point

If one breaks this symmetry—say, by putting graphene on a substrate that treats the *A* and *B* sublattices differently—the bands immediately repel each other, and a bandgap opens up

This is how people use engineering to alter quantum states

I. Crystal Symmetry

Time-Reversal Symmetry and Spintronics Implications

A band structure diagram showing a spin-up band (red) and a spin-down band (blue). In a non-magnetic crystal, they overlap perfectly. In a ferromagnetic crystal, they shift away from each other along the energy axis

The Time-Reversal Operator ($\widehat{T}$): Flips the direction of time ($t \rightarrow -t$). In quantum mechanics, it reverses both the momentum vector ($k \rightarrow -k$) and the electron spin ($\uparrow \rightarrow \downarrow$)

Kramer's Degeneracy Theorem: In any system with an odd number of electrons under time-reversal symmetry (and no external magnetic fields), every energy state is at least two-fold degenerate $E_n(k,\uparrow) = E_n(-k,\downarrow)$

Inversion + Time Reversal (PT Symmetry): If a crystal has both Spatial Inversion ($\widehat{P}$) and Time-Reversal ($\widehat{T}$) symmetry, the spin-up and spin-down bands are completely locked together at every single k point $E_n(k,\uparrow) = E_n(k,\downarrow)$

Breaking Symmetry for Spintronics: To split the spins and generate spin currents or Spin-Orbit Torques (SOT), one must deliberately introduce magnetic elements or surfaces to break these symmetry ties

I. Crystal Symmetry

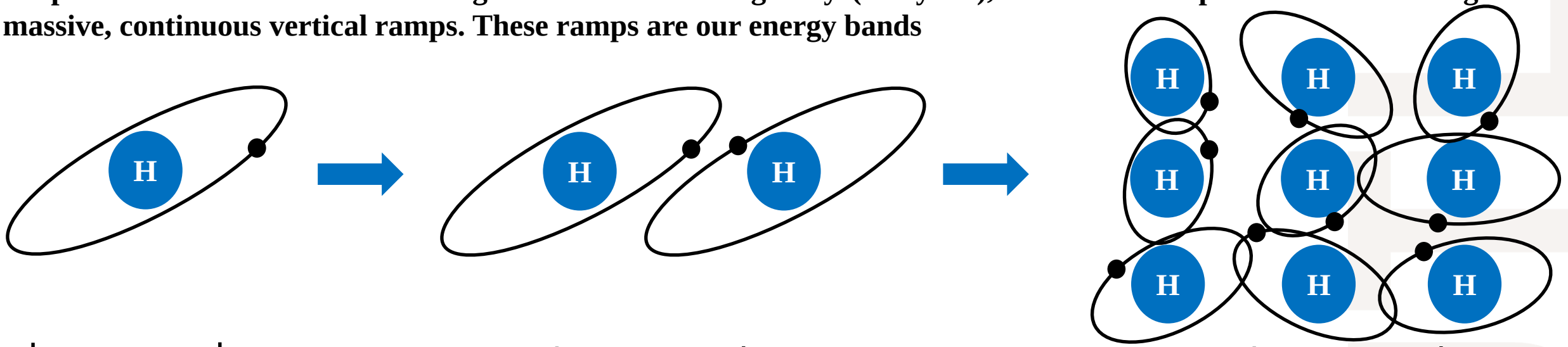
For those who will be running spintronics calculations, this slide is your bible. If the material has an inversion center and is non-magnetic, those spin-up and spin-down bands will look like a single line in the band plot.

If one is trying to calculate unconventional Spin-Orbit Torques or Orbital Torques, one must intentionally look for materials that lack inversion symmetry—such as Janus 2D materials or complex Kagome lattices like Mn_3Sn—so that the spin states can separate in momentum space

II. Band Structure

The Origin of Bands: From Isolated Atoms to Periodic Solids

Think of isolated atoms as lonely houses. The electrons live on neat, discrete floors (atomic levels 1*s*, 2*s*, 2*p*). But when we pack billions of these houses together to build a mega-city (a crystal), the walls collapse. The floors merge into massive, continuous vertical ramps. These ramps are our energy bands



Bonding and anti-bonding bands

A couple of bands

II. Band Structure

♦ **The Pauli Exclusion Principle: No two electrons in an interacting system can occupy the identical quantum state**

♦ **Orbital Splitting: When *N* atoms assemble to form a crystal lattice, their valence electron wave functions overlap, forcing each discrete atomic orbital to split into *N* closely spaced molecular orbitals**

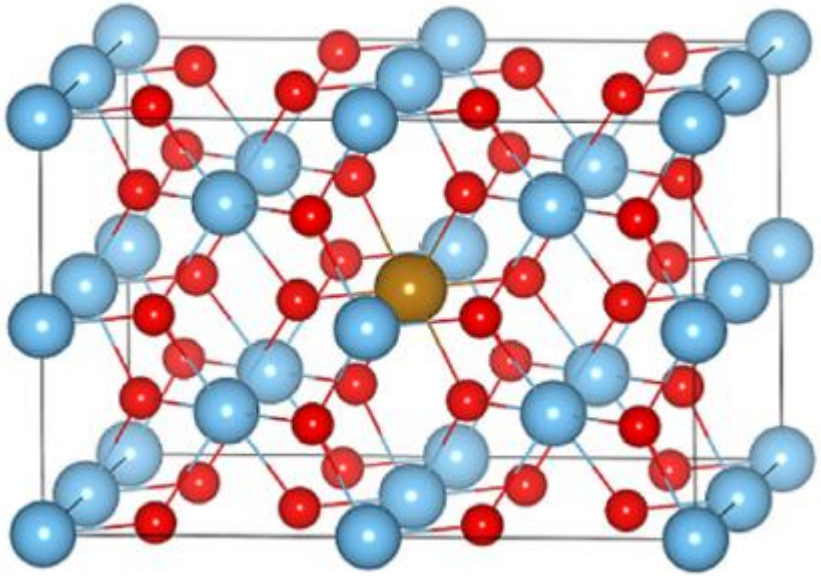
♦ **Band Formation: In a macroscopic crystal ($N \approx 10^{23}$), these split states are so infinitely dense that they form a continuous domain of allowed energies—an Energy Band**

♦ **Forbidden Gaps: The regions between atomic shells where no quantum states can exist become Band Gaps**

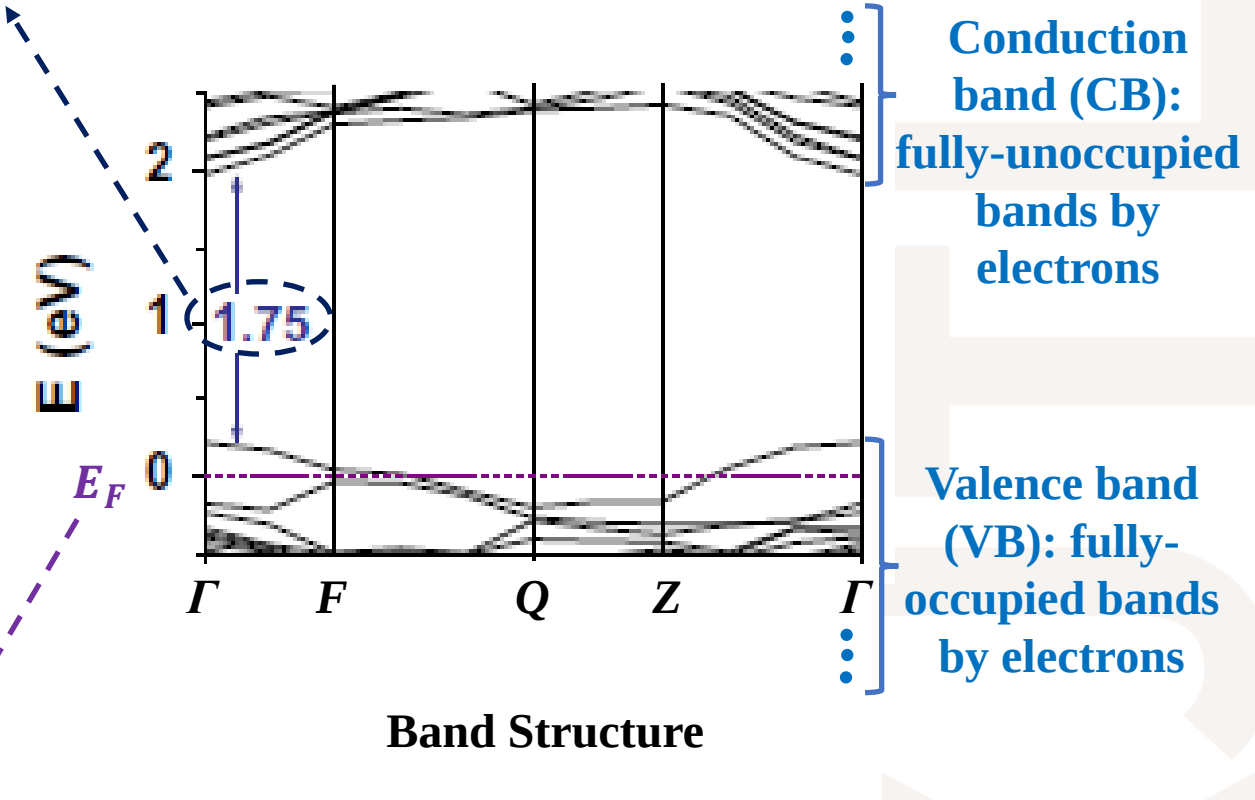
II. Band Structure

Semiconductor: 0 eV < Gap value ≤ 5 eV

Band Gap value: Energy difference between minimum of conduction band and maximum valence band



Yttrium doped TiO_2 with Rutile phase



Band Structure

Fermi Energy Level (E_F): border separating fully-occupied bands and fully unoccupied bands by electrons

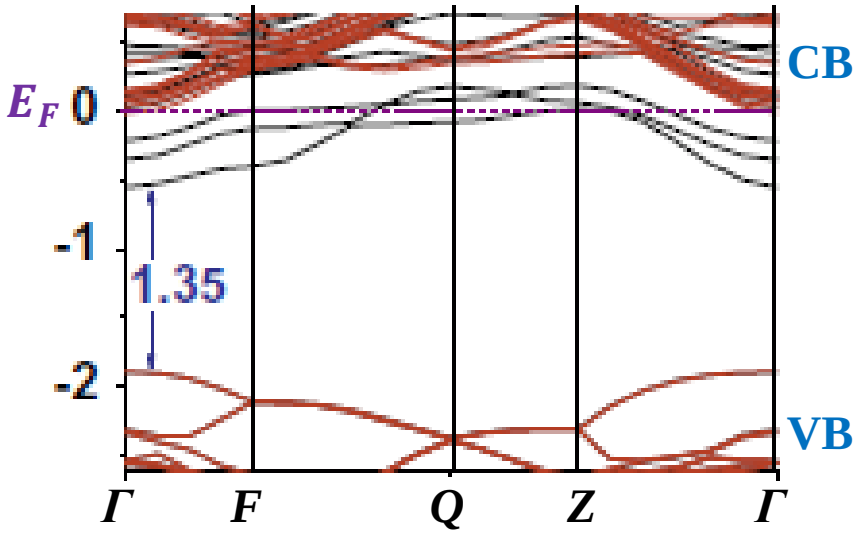
Kenan Song, Xiaoping Han and Guosheng Shao, Electronic properties of rutile TiO_2 doped with 4*d* transition metals: First-principles study, Journal of Alloys and Compounds 551 (2013) 118-124

II. Band Structure

***n*-type Semiconductor: electron is the main carrier and E_f is close to minimum of Conduction Band**

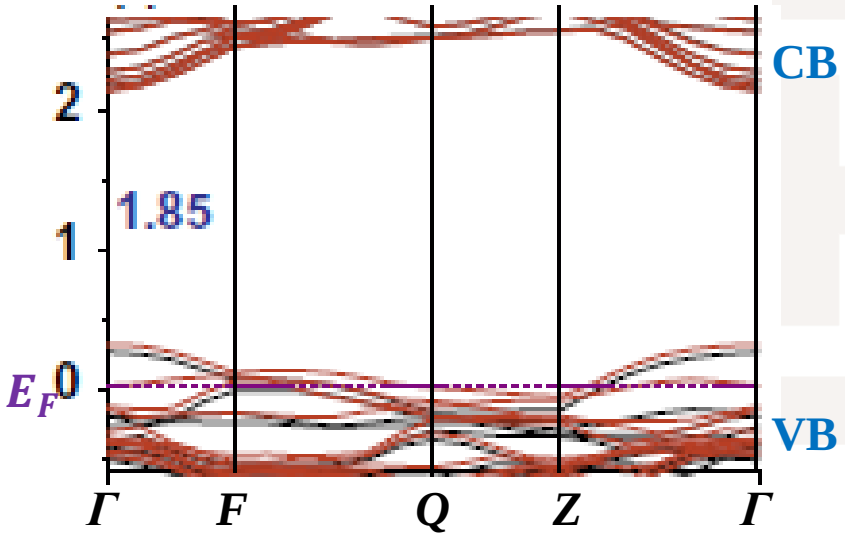
***p*-type Semiconductor: hole is the main carrier and E_f is close to minimum of Valence Band**

***n*-type Semiconductor**



Molybdenum doped TiO_2 with Rutile phase

***p*-type Semiconductor**

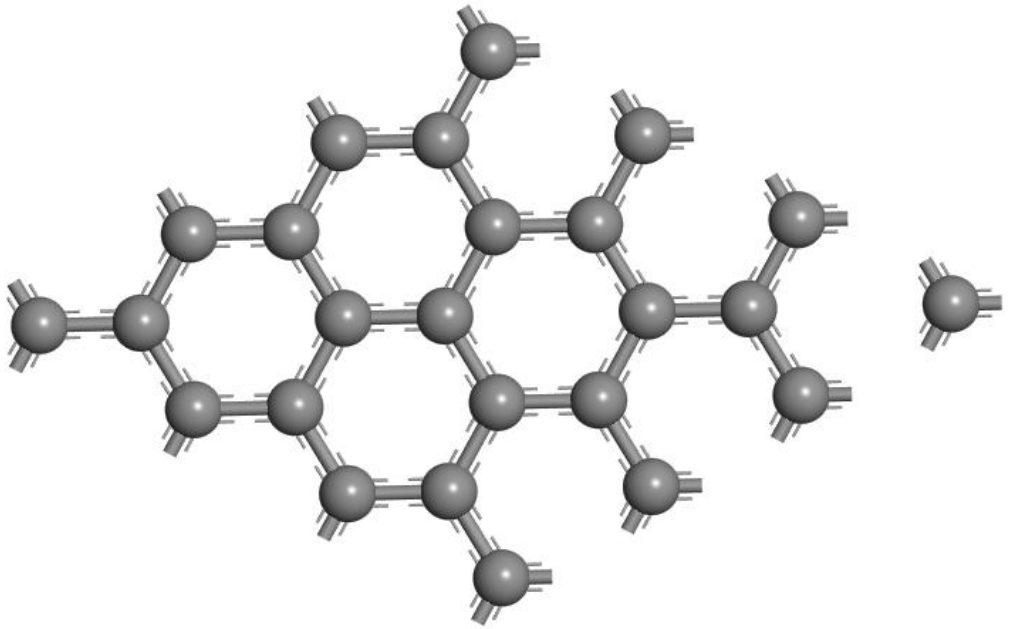


Cadmium doped TiO_2 with Rutile phase

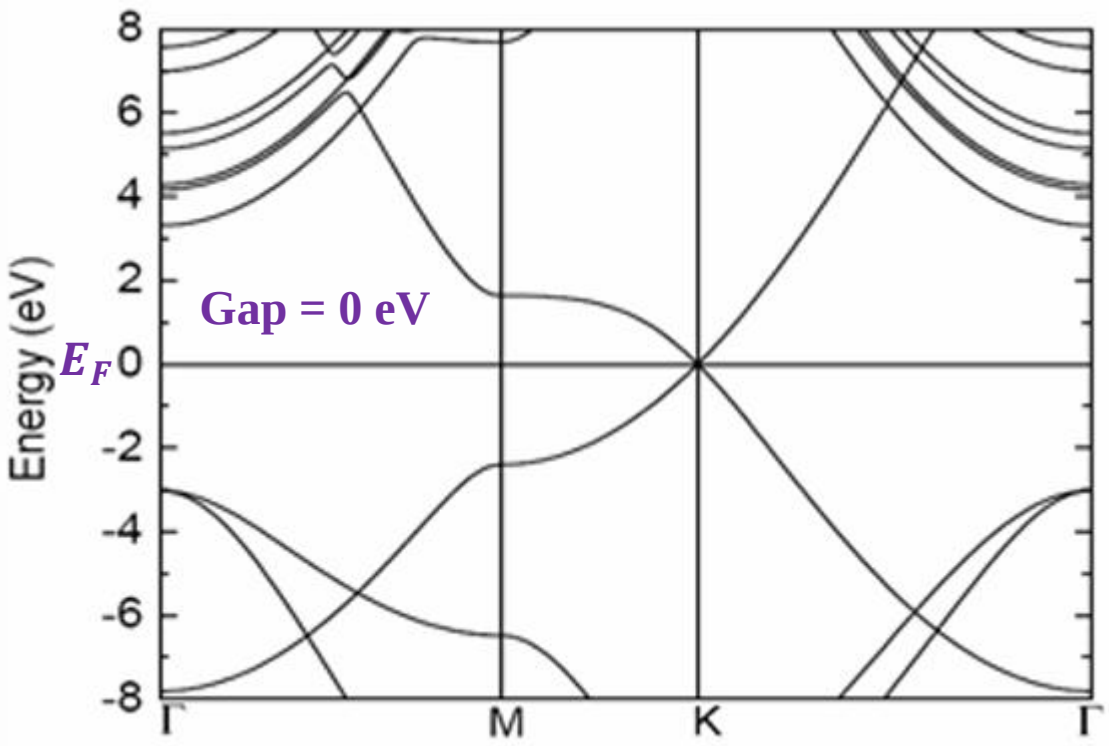
Kenan Song, Xiaoping Han and Guosheng Shao, Electronic properties of rutile TiO_2 doped with 4*d* transition metals: First-principles study, Journal of Alloys and Compounds 551 (2013) 118-124

II. Band Structure

Conductor: Gap value = 0 eV



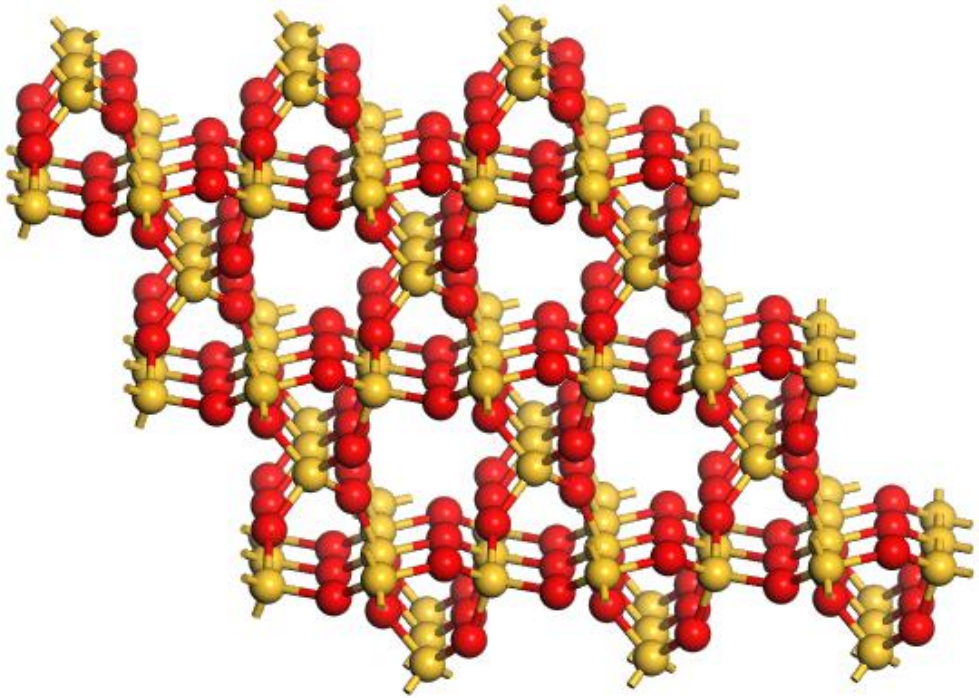
Pristine Graphene



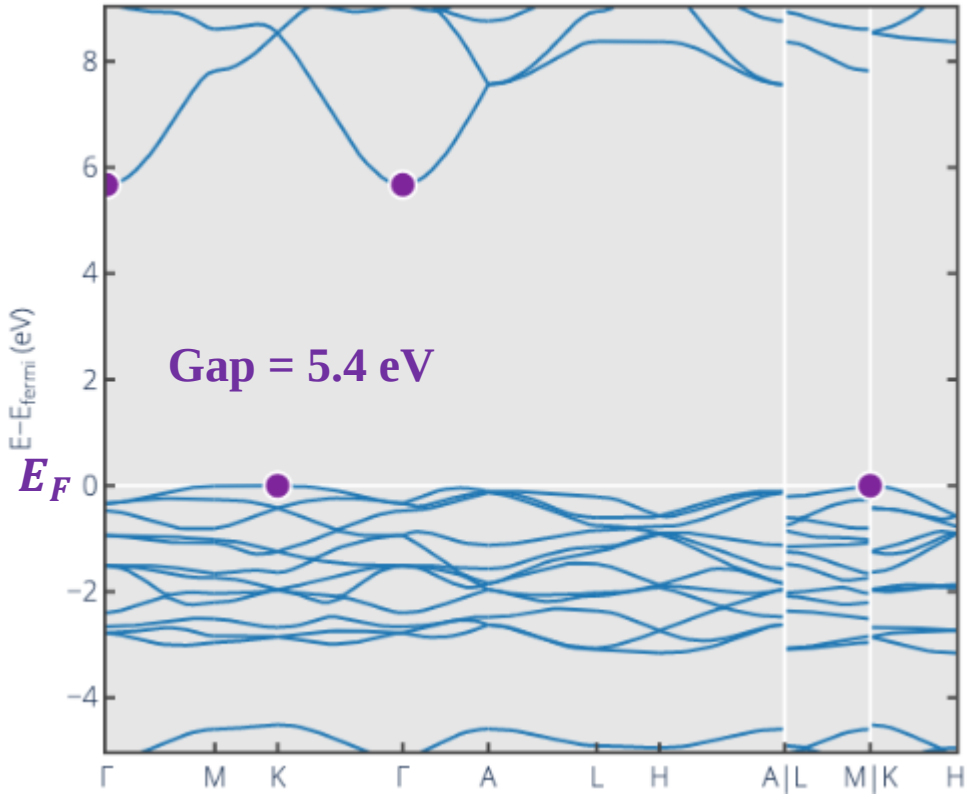
Harihar Behera and Gautam Mukhopadhyay, Structural and Electronic Properties of Graphene and Silicene: An FP-(L)APW+lo Study, AIP Conf. Proc. 1313, 152-155 (2010)

II. Band Structure

Insulator: Gap value > 5 eV



Pristine SiO_2



II. Band Structure

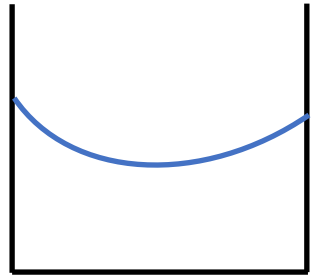
Curvature and Effective Mass (m^*): The local curvature dictates how 'heavy' the electron behaves

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k^2}$$



Carrier Mobility: none

Full Isolated band: electron on this band cannot move onto other band



Carrier Mobility: good

Not Isolated band: electron on this band can move onto other band

II. Band Structure

The Quantum Language—Bloch's Theorem

Electrons as Waves: Bloch Theorem and Quasi-Momentum

A mathematical plot showing a plane wave $e^{ik\cdot r}$ modulated by a periodic function u_{nk} that peaks exactly at every atomic nucleus site

The Schrodinger Equation in a Crystal: The potential energy $V(r)$ is perfectly periodic: $V(r+R)=V(r)$, where R is the real-space lattice vector

Bloch's Theorem: The eigenstates of this periodic Hamiltonian must take the form of a modulated plane wave

$$\Psi_{nk}(r) = u_{nk}(r)e^{ik\cdot r}$$

$e^{ik\cdot r}$: The free-electron plane wave carrier
$u_{nk}(r)$: A periodic function possessing the full symmetry of the crystal lattice: $u_{nk}(r+R)=u_{nk}(r)$
n: The Band Index (1st band, 2nd band, *etc.*)
The Wave Vector k: Acts as the quantum label (or coordinate) inside the First Brillouin Zone

II. Band Structure

s — only one orbital

p — p_x, p_y and p_z three orbitals

d — d_{xy}, d_{xz}, d_{yz}, d_{z^2} and $d_{x^2-y^2}$ five orbitals

Example: Ga_2O_3

Basis: Ga7.0-s2p2d2
O6.0-s2p2d1

Number of Ga: 64
Number of O: 96

Electron Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

Main Quantum Number (n)

Orbital Angular Momentum Number (l)

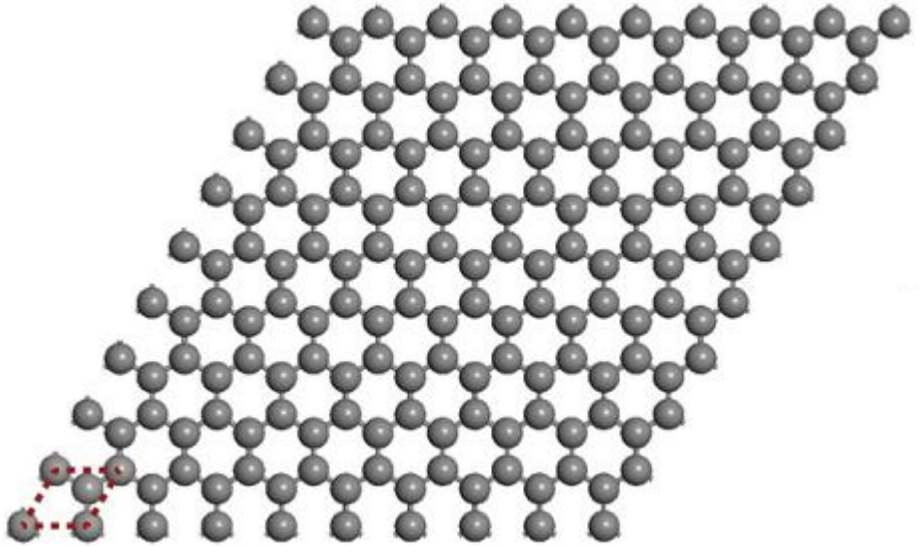
Electron Number

Ga: 2×1+2×3+2×5 = 19
O: 2×1+2×3+1×5 = 13

Band for Ga: 19×64 = 1216
Band for O: 13×96 = 1248

Total Number of Band: 1216+1248 = 2426

II. Band Structure



Pristine Graphene

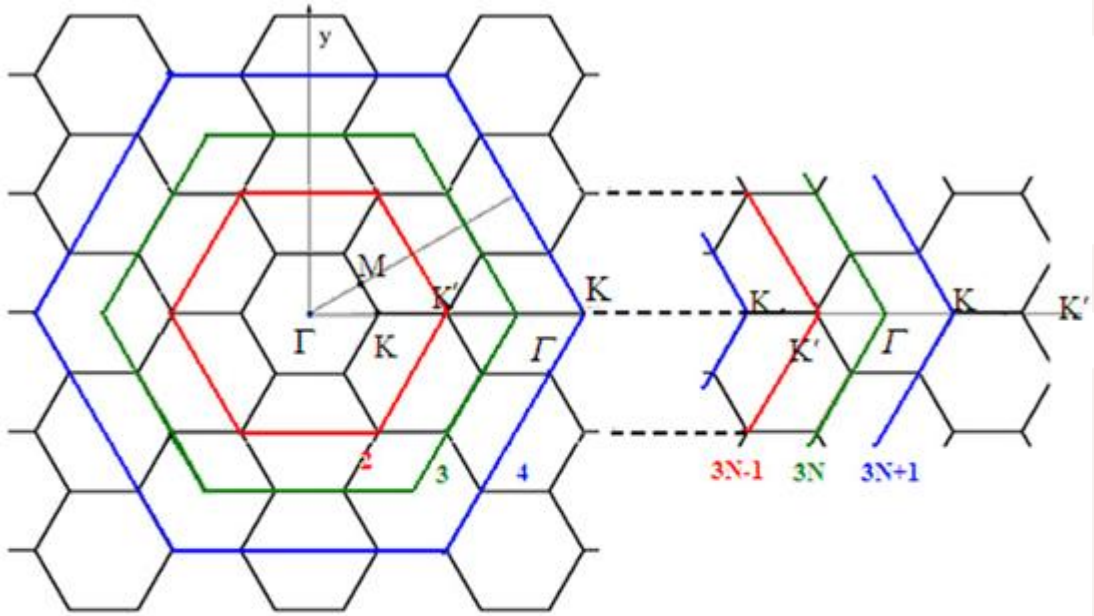
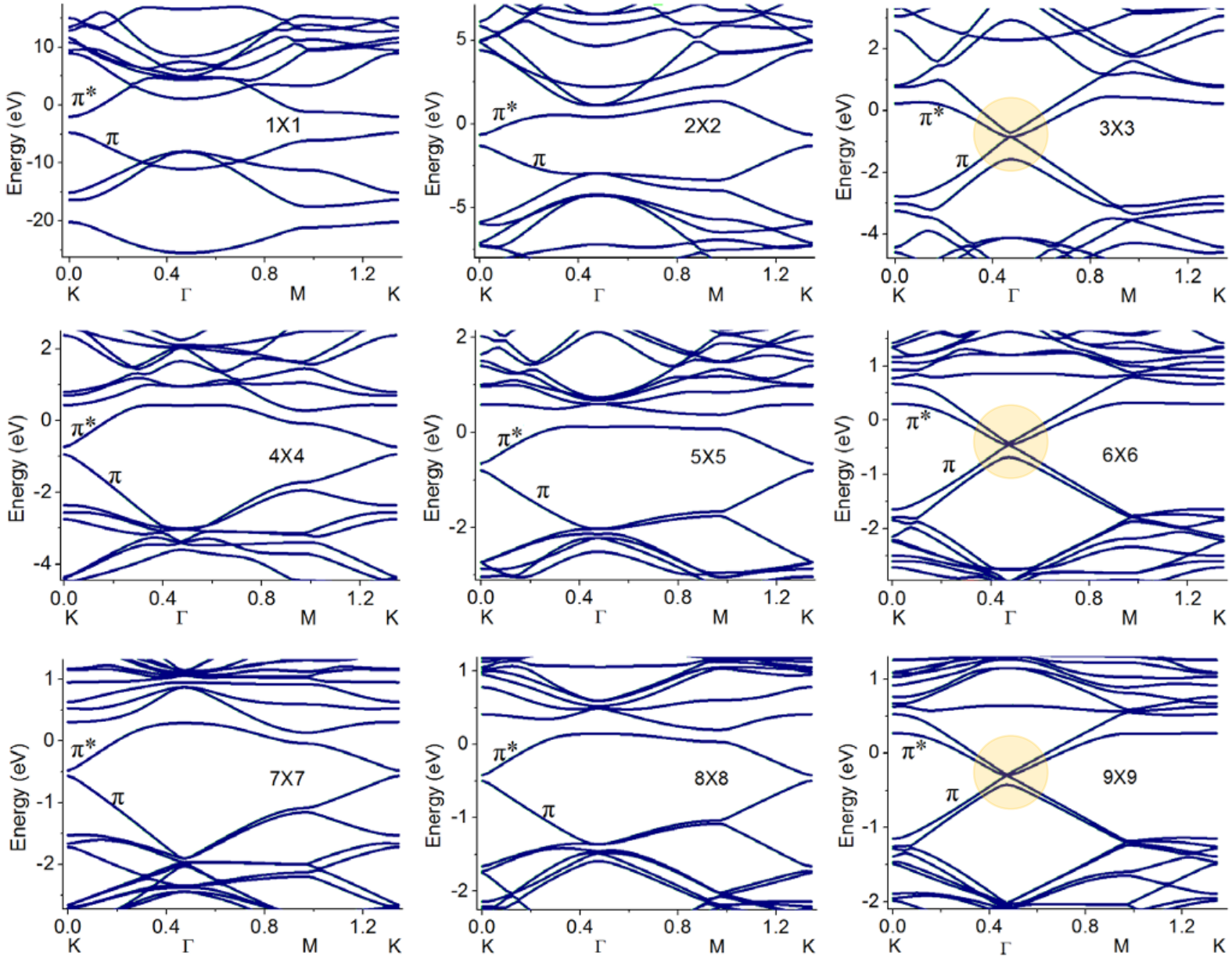


Figure 4. Band folding of graphene. The black lines represent the first Brillouin zone of the supercell. The red, green and blue lines are the Brillouin zones of primitive cell corresponding to the supercells of $(3N-1)\times(3N-1)$, $3N\times 3N$ and $(3N+1)\times(3N+1)$, respectively.

Y. C. Zhou, H. L. Zhang and W. Q. Deng, A 3*N* rule for the electronic properties of doped graphene, Nanotechnology 24 (2013) 225705

II. Band Structure

3*N*×3*N* supercell gives the Dirac point folded onto point in first BZ



Y. C. Zhou, H. L. Zhang and W. Q. Deng, A 3*N* rule for the electronic properties of doped graphene, Nanotechnology 24 (2013) 225705



That's all for this session

Thank you for your attention and great participation